Stem-cell Research

All of us would like to see our world as a haven, full of people free from any sort of impairment. But at what cost? Medical science has its eyes set on the highly controversial topic of stem-cell research. Current work on this subject shows promising results, but there is a catch. The only place stem cells are available is within a human embryo and removing these cells leads to the death of the embryo. Stem cells are known as 'blank cells' as they can be easily transplanted and specialised into any kind of cell for the body—as muscles, blood or brain cells. For example, a stem cell can be easily inserted into the heart in the event of heart disease and can heal the area that has been damaged. Stem-cell research has the potential to literally cure several diseases such as Parkinson's, diabetes, deafness and stammering.

Now, newer research quells the controversy of this line of study by using adult human stem cells extracted from the umbilical cord, fat tissue and bone marrow as opposed to using embryonic cells. One key advantage to using adult stem cells is that the chance of cell rejection is much less. For example, stem cells found in bone marrow can be used to heal another problem in your body without the fear of a stem cell being thrown out.

This technology is still in its infancy and is being tested (successfully) on animals. Researchers also forecast that practical treatments from stem cell study are still a few years away.

When people with normal hearing actually hear, sound enters the auditory canal and triggers hair cells in the cochlea, which is a spiral shaped tube filled with fluid. The hairs then send information to our hearing nerve and routes the other information to the brain. With the implant, a very small microphone picks up sound and sends it across to its processor, which is situated outside the ear. The processor then sends signals through the skin directly to the hearing nerve sidestepping the hairs that no longer work.

Hearing aids are tools commonly used by the deaf to amplify sounds. The depth and direction from sound allows those with normal hearing to determine where the sound originates. A hearing aid cannot help its user in determining the direction from which a sound originates. A device being developed by Swedish researchers might change this. Using a technology used in locating submarines in naval warfare, the team has devised a system that will allow a person with partial hearing loss to identify the direction of a sound's source. This system is designed to integrate with a pair of sunglasses. A pair of microphones pick up sound and an attached digital signal processor (DSP) calculates the route of the sound, similar to what a human ear does. Two small vibrating pads mounted on soft pads are attached to the temples and pulsate on the left or the right depending upon the bearing of the sound wave. Certain Morse code-like pulses are used to determine sounds coming from different areas.

Another famous technology with the deaf community is the Internet—it's a barrier-free environment and lets the deaf remain at the same level with normal people.

Technology steps in

Those suffering from debilitating diseases or with missing limbs can take respite in what today's expertise can offer. Getting mobile means making use of specially designed scooters, keeping in mind the needs of the immobile, such as ease of access and control. Motorised wheelchairs with computerised interfaces are another bi-product of what technology can do. For added comfort, researchers are in the midst of developing a sensory seat which picks up subtle changes in movements of the occupant and changes the seating position accordingly to create a more cozy and correct seating posture. Doctors have noted that posture is of key significance to the proper flow of blood within the body.

Physical immobility as a result of a disease or missing or paralysed limbs can possibly see technology create motion, not through physical means, but from the power of the mind. That's what scientists are edging closer to with the use of robotic limbs. From the realms of science fiction, successful tests have shown that the brain power of a monkey can control the movement of an artificial robotic arm—even across continents. Developers are confident that an arm or leg made of steel controlled by the mind will have similar dexterity as natural human move-

Superhuman Achievements

One particular individual's achievements fed immense courage to those who endure impairments on a daily basis. Well known astrophysicist Stephen Hawking suffers from Amyotrophic Lateral Sclerosis (ALS or Lou Gehrig's disease), a disease which leads to paralysis by slow weakening of the muscles. In spite of his incapacity to move or speak, Hawking is often touted as the greatest genius of modern history since Einstein. A total understanding of the universe will be his nirvana. Technology remains in true favour not for Hawking, but for us by making it possible for the astrophysicist to share his research with fellow scientists and the public as well.

His inability to speak led Hawking to use Words+, a software package that can predict and complete words of sentences from those with disabilities like ALS. The program also converts text to speech where Hawking jokes, "The only trouble is that it gives me an American accent." E Z Keys also doubles as an environmental-control unit capable of switching on/off appliances in the house.

In unison with E Z Keys, Hawking's second helping of Assistive Technology is a small portable computer and a speech synthesiser strapped to his wheelchair. Using technology, Hawking has been able to write countless scientific articles, deliver hundreds of lectures and write a few books such as his best-seller, *A Brief History of Time*. Without the presence of this technology, solid breakthroughs in quantum physics would have otherwise been impossible.